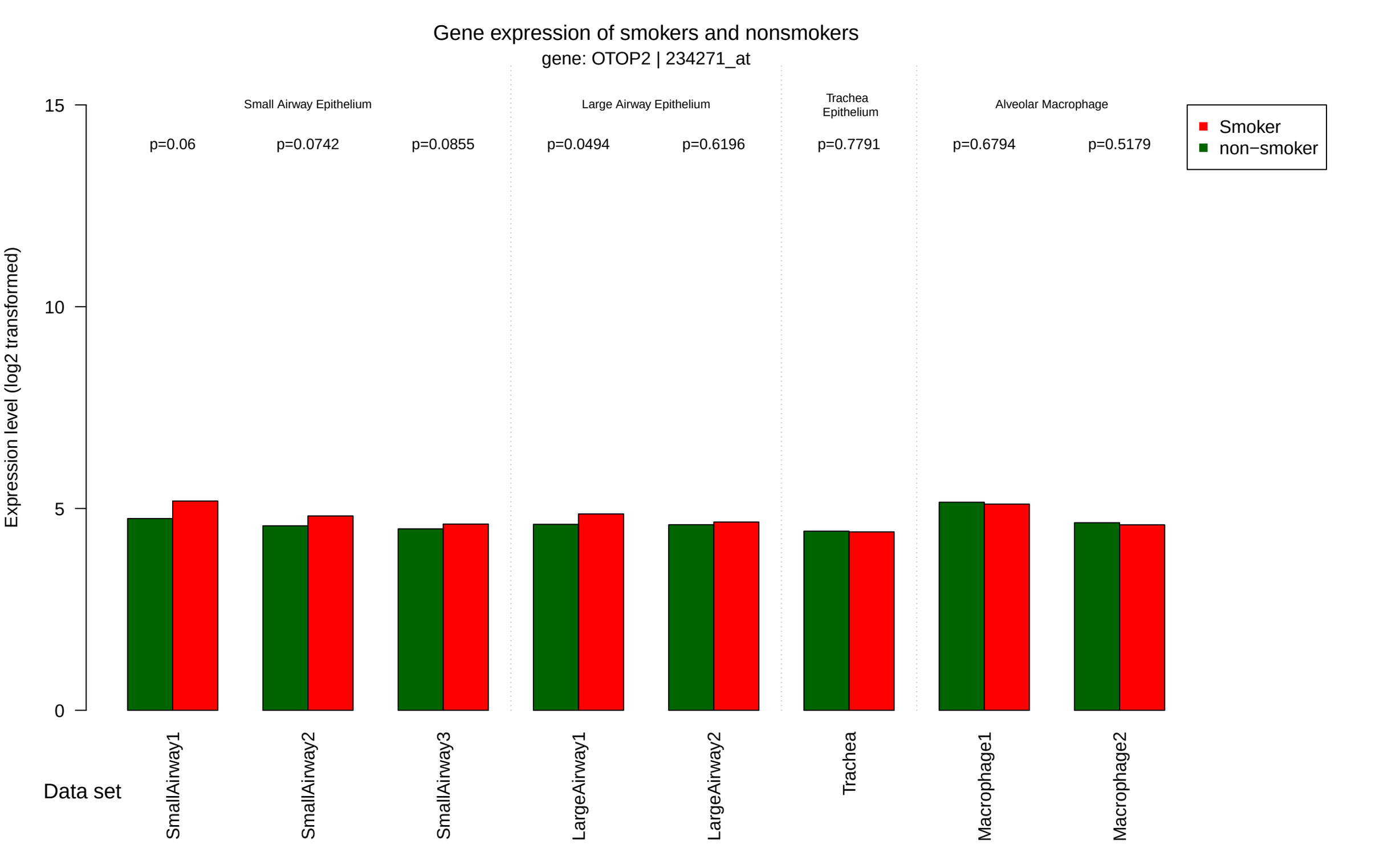
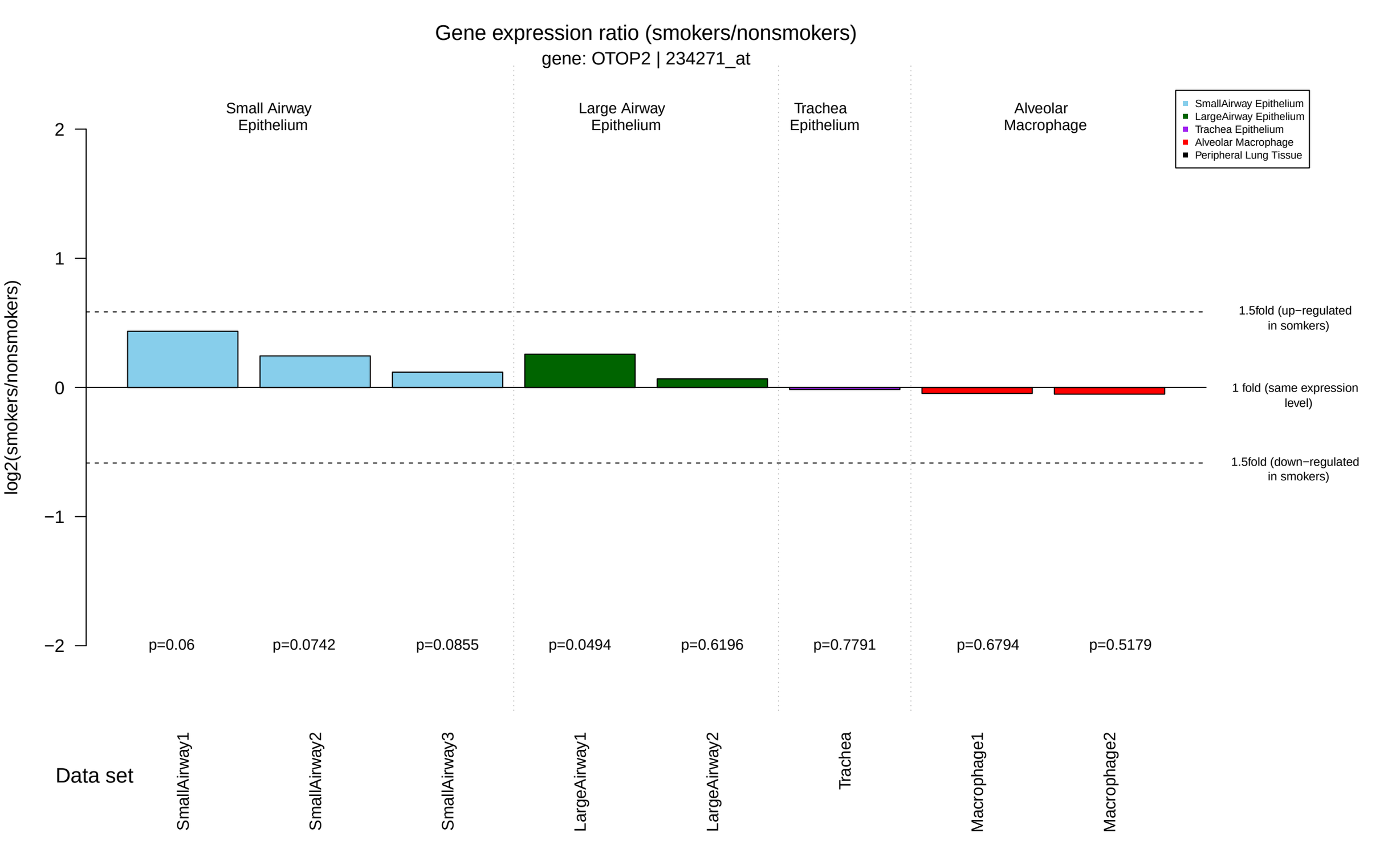
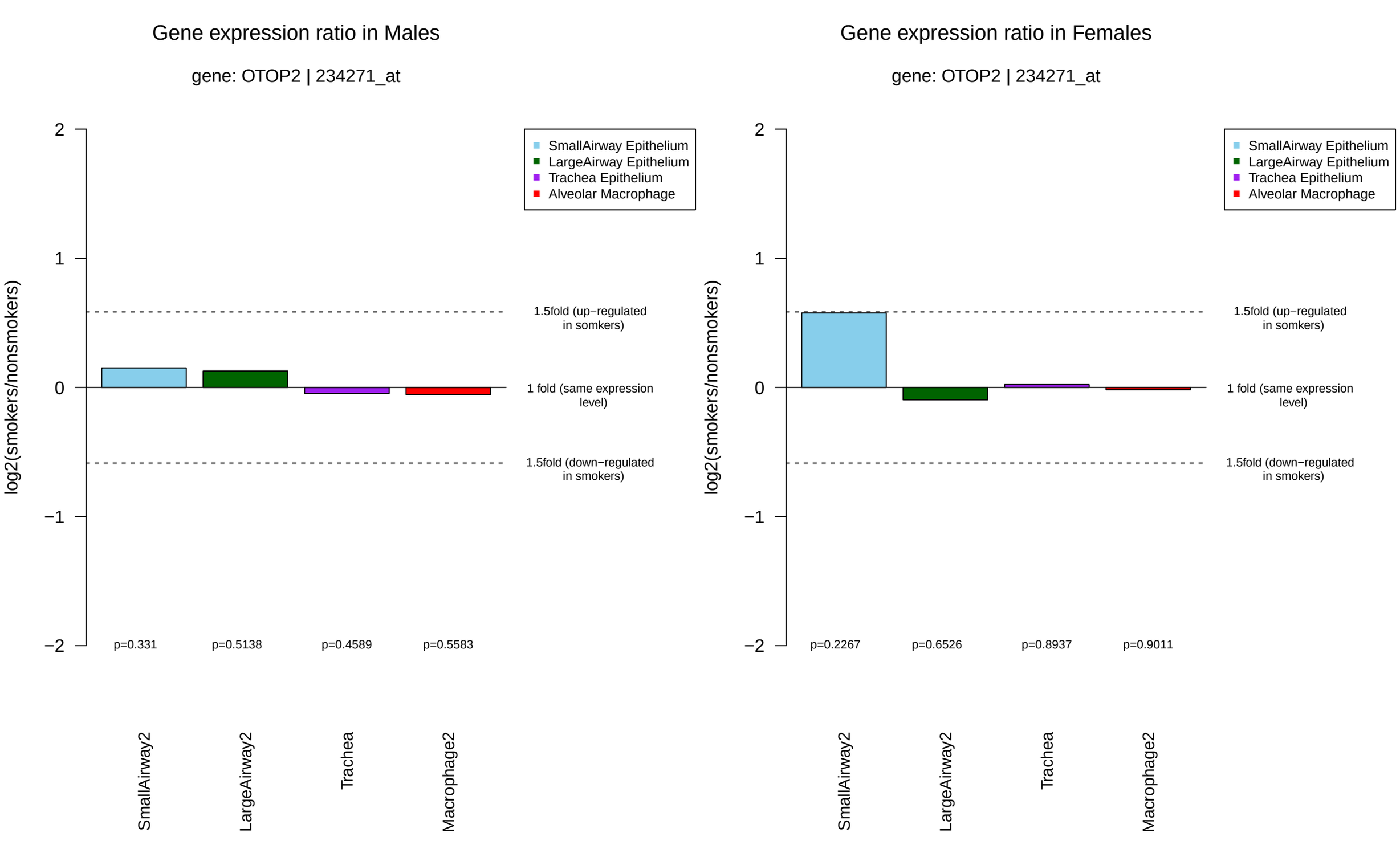
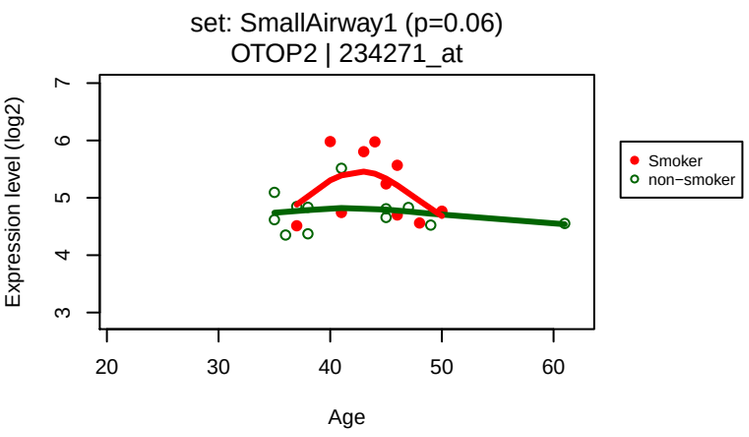
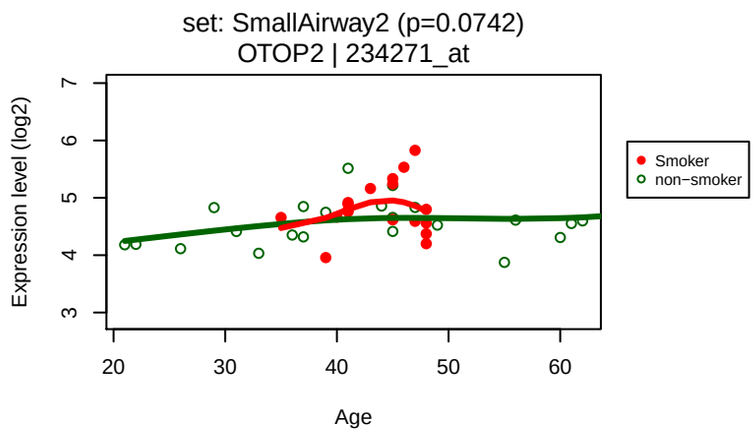
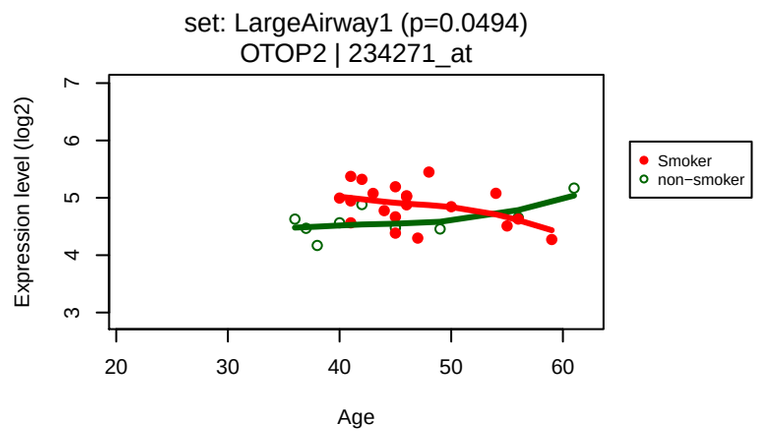

Gene expression of smokers and nonsmokers
gene: OTOP2 | 234271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.06
p=0.0742
p=0.0855
p=0.0494
p=0.6196
p=0.7791
p=0.6794
p=0.5179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

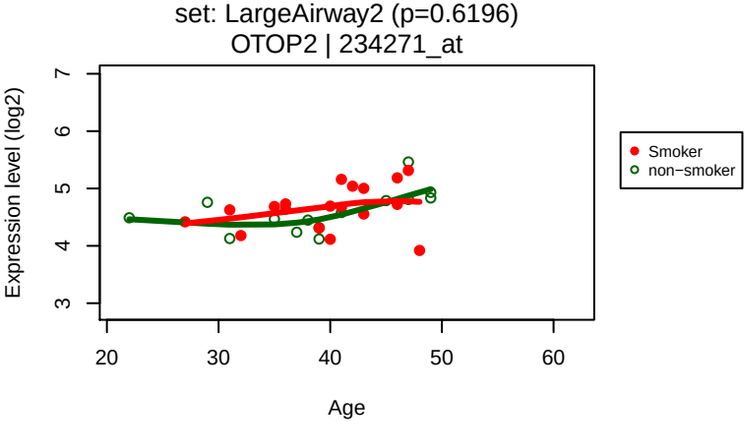

Gene expression ratio (smokers/nonsmokers)
gene: OTOP2 | 234271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.06
p=0.0742
p=0.0855
p=0.0494
p=0.6196
p=0.7791
p=0.6794
p=0.5179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

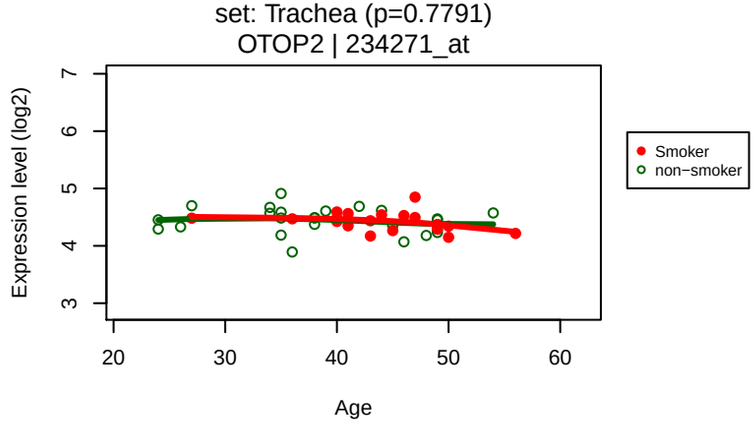

Gene expression ratio in Males
gene: OTOP2 | 234271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.331
p=0.5138
p=0.4589
p=0.5583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OTOP2 | 234271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2267
p=0.6526
p=0.8937
p=0.9011
SmallAirway2
LargeAirway2
Trachea
Macrophage2

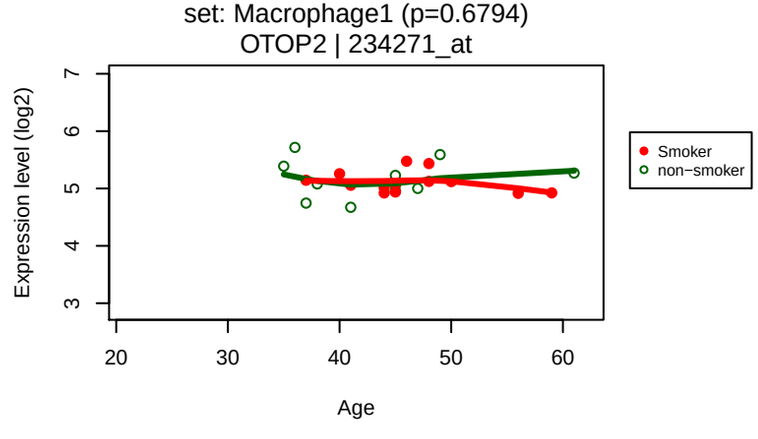

set: SmallAirway1 (p=0.06)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker

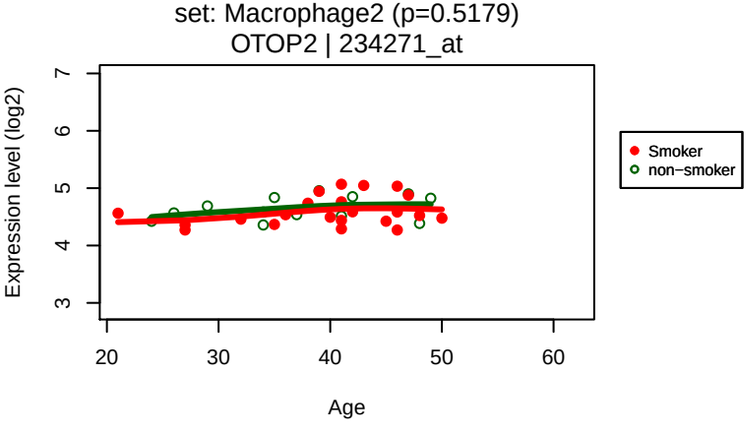

set: SmallAirway2 (p=0.0742)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker

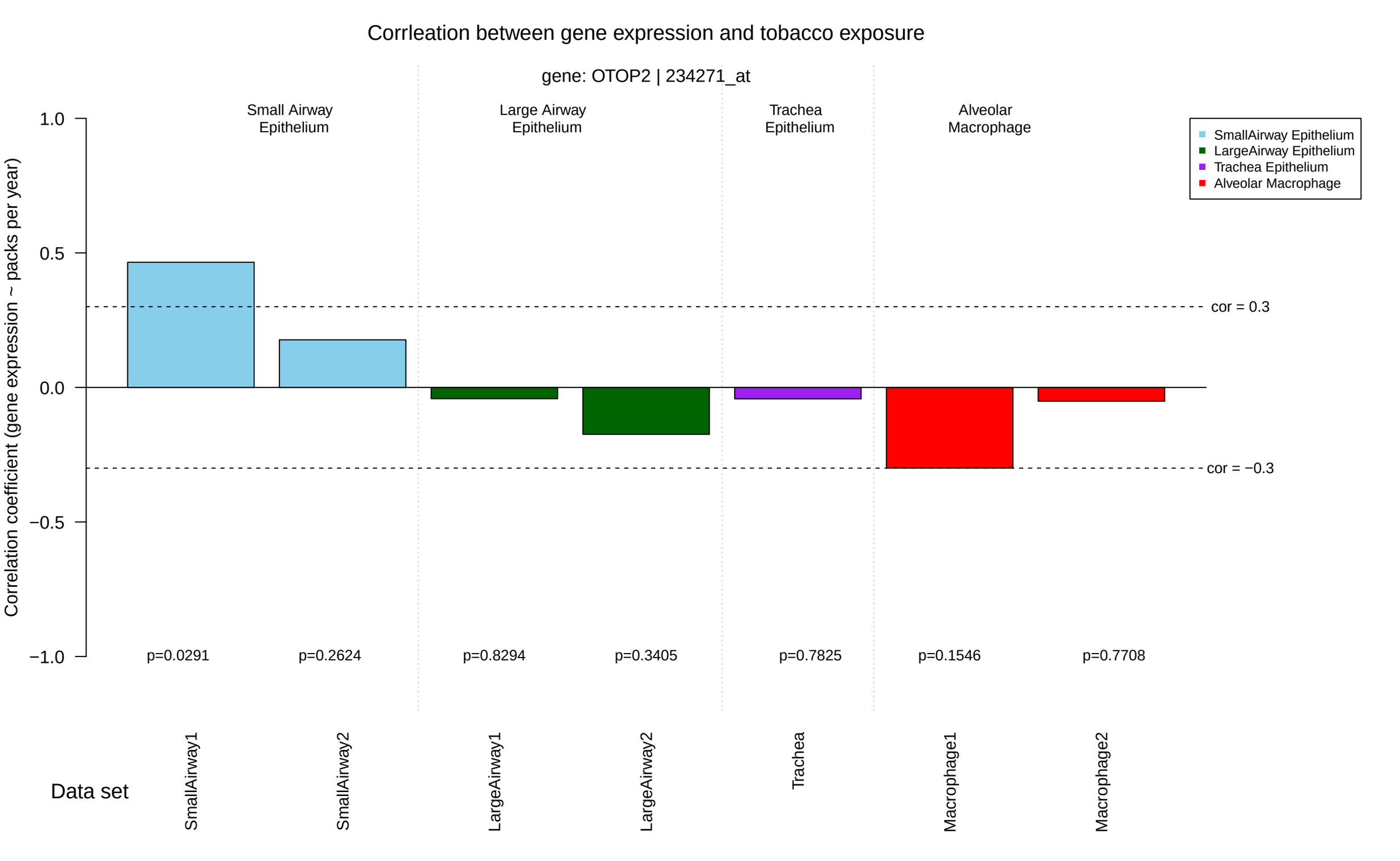

set: LargeAirway1 (p=0.0494)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6196)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7791)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6794)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5179)
OTOP2 | 234271_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OTOP2 | 234271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0291
p=0.2624
p=0.8294
p=0.3405
p=0.7825
p=0.1546
p=0.7708
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2